

Supplementary Information

A Multi-Vendor Corpus of Reference Low-Dose CT Reconstructions with Calibrated Per-Pixel Uncertainty and Signal-Equivalence Credentials

This supplement records the configuration needed to regenerate the corpus bit-for-bit (within IEEE-754 tolerance) from the deposited Apache-2.0 RunBundle. **Table S1** lists the reconstruction and training hyperparameters; **Table S2** lists the determinism settings and pinned software versions; **Table S3** discloses training compute. Entries marked **[TODO:]** are fixed when the Phase-3 models are trained and the corpus is deposited.

S1 Hyperparameters

S2 Reproducibility: determinism and software pins

S3 Training-compute disclosure

Group	Hyperparameter	Value
<i>Unrolled reconstruction</i>		
	Iterations K	10
	Warm start	TV-regularized FBP (TV-FBP)
	Data-consistency step size τ	learned scalar, init $0.5/\ \mathcal{R}\ _2^2$
	Reconstruction loss	mean absolute error (L1) on HU
	Forward / adjoint operator	in-repo <code>pwm_ldct_recon.physics</code> (parallel beam; interface-compatible with <code>pwm_cor</code>)
	Projection geometry (n_{views} , n_{dets})	[TODO: geometry]
	FBP filter	[TODO: filter (e.g. Ram-Lak / Hann)]
<i>Denoiser f_θ (weights shared across iterations)</i>		
	Architecture	U-Net, 4 encoder / 4 decoder stages
	Channels per stage	(32, 64, 128, 256)
	Activation / normalisation	GELU / GroupNorm
	Upsampling	bilinear
	Parameters per ensemble member	≈ 4 M
<i>Optimisation</i>		
	Optimiser	AdamW
	Peak learning rate	[TODO: lr]
	Weight decay	[TODO: wd]
	LR schedule	cosine, 5% linear warmup
	Training length (epochs / steps)	[TODO: epochs]
	Batch size	4
	Input	full 512×512 axial slice (2D)
<i>Training data</i>		
	Dose levels r	{0.10, 0.25, 0.50}
	Per-minibatch dose-mix weights	(0.4, 0.4, 0.2) for $r = (0.10, 0.25, 0.50)$
	Augmentation	flip $p=0.5$; rotation $[-5^\circ, 5^\circ]$; window-level shift $[-50, 50]$ HU
	Train / val / test sizes (patients)	[TODO: N / N / N]
<i>Ensemble</i>		
	Members M	5
	Member seeds	{42, 43, 44, 45, 46}

Table S1: Reconstruction, denoiser, optimisation, data, and ensemble hyperparameters. Fixed values mirror the Methods; **[TODO:]** entries are set at training time.

Group	Setting / package	Value
<i>Seeding and determinism</i>		
	Seeded RNG streams	Python <code>random</code> , NumPy, PyTorch (CPU + CUDA), DataLoader workers
	<code>cuda.deterministic</code>	True
	<code>cuda.benchmark</code>	False
	<code>torch.use_deterministic_algorithms</code>	False (global enable is incompatible with the differentiable <code>grid_sample</code> in t)
	<code>PYTHONHASHSEED</code>	0 (set in the RunBundle image)
	<code>OMP_NUM_THREADS</code> / <code>MKL_NUM_THREADS</code>	1 / 1 (set in the RunBundle image)
	Per-member seed $\rightarrow$ weights + data order	{42, ..., 46} (Table S1)
<i>Pinned software versions</i>		
	Python	3.12 (RunBundle base <code>python:3.12-slim</code> ; exact patch pinned by the base-in
	PyTorch	[TODO: x.y.z]
	CUDA / cuDNN	[TODO: x.y / x.y]
	NumPy	[TODO: x.y.z]
	SciPy	[TODO: x.y.z]
	<code>pwm_ldct_recon</code> (in-repo Radon operator)	[TODO: repository commit / image digest]
	<code>pwm_dose_equivalence</code>	0.2.2
	<code>nnDetection</code> (frozen task detector)	[TODO: release / commit]
	<code>nibabel</code>	[TODO: x.y.z]
	Docker base image digest	[TODO: sha256:...]
<i>Reproduction tolerance</i>		
	Bit-identical within	IEEE-754 FP tolerance on equivalent hardware
	Framework hash (credentials)	sha256:b366f51c... (<code>pwm_dose_equivalence</code>)

Table S2: Determinism settings, pinned software versions, and the framework hash credentials are issued under. These plus the RunBundle permit bit-reproduction of every record. The framework hash in full is `sha256:b366f51c11a8fbd7c7b60f1f977d42a10a8bdaa55521b43de0c428afc289fb58`, the SHA-256 of `pwm_dose_equivalence`’s `FRAMEWORK_SPEC` (v0.2.2); a credential carrying any other hash was issued under an incompatible framework version and is out of scope for this corpus. The exact resolved versions of the floating dependencies (PyTorch, NumPy, SciPy, nibabel) are captured by `pip freeze` into a lockfile deposited with the RunBundle at build time, since the Dockerfile constrains but does not exactly pin them.

Quantity	Value
GPU model	NVIDIA A100 (single)
GPU-hours per ensemble member	[TODO: h]
Ensemble members trained	5 (reference) + cross-vendor leave-one-out variants
Total training GPU-hours (all models)	[TODO: h]
Inference time per slice (5-model ensemble)	[TODO: s on one A100]
Peak GPU memory (training)	[TODO: GB]
Estimated energy / CO ₂ e	[TODO: kWh / kg (optional)]

Table S3: Training-compute disclosure for the released models.